card, and communications between the various chipsets on the motherboard—the hard drive with its 10 ms seek times (on average), seem prehistoric; a snail's pace in a jet age.

To alleviate this bottleneck, Vista introduces another layer of storage between the fast (yet expensive) RAM, and the slow (yet inexpensive) hard drive. This layer is that of the Flash memory: Flash is cheaper than RAM, but faster than the typical hard drive by a factor of 10, thus it makes for a good in-between solution.

Vista thus makes use of any Flash device as a system cache, in lieu of the traditional hard drive cache. The idea is that the Flash cache is given priority if available and thus system bottlenecks are much reduced.

This feature of Vista falls under the banner of ReadyBoost. ReadyBoost is of course only used if there is a Flash device present—while some motherboard will soon sport minimal Flash storage devices in the near future, currently this would largely be limited to your music player or to a USB drive.

So how does ReadyBoost work? Firstly, the Flash device should be between 256 MB and 32 GB in size, with a transfer rate of 2.5 MB/s or higher for random 4 KB reads, and 1.75 MB/s or higher for random 512 KB writes. When you insert such a device, Vista asks if you would like to dedicate up to 4 GB of the disk for the purpose of data caching. Note that the 4 GB is purely for the benefit of FAT32 devices... if you agree, a caching file named ReadyBoost.sfcache is created in the root of the device. SuperFetch then repopulates this cache in the background and then triggers the Ecache.sys device driver to intercept all reads and writes to local hard disk volumes. The data is then written into the cache file with a compression ratio of 2:1. Thus the 4 GB cache can typically store up to 8 GB of actual data. The data stored is also encrypted using Advanced Encryption Standard (AES). The key for encryption is randomly generated per-session in order to guaran-